INCH-POUND
MIL-DTL-23765/3B(SH)
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SUPERSEDING
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11 March 1986

DETAIL SPECIFICATION SHEET

ELECTRODES AND RODS - WELDING, BARE, SOLID COPPER ALLOY

This specification is approved for use by the Naval Sea System Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-23765.

1. SCOPE

1.1 Scope. This specification covers solid bare copper alloy electrodes and cut-length rods for use with the gas metal-arc welding (GMAW) and gas tungsten-arc welding (GTAW) processes.

1.2 Classification. Electrodes and rods are provided in the types specified in table I and in forms 3a, 3b, 3c, 3d, 4, and 6 with sizes in accordance with MIL-DTL-23765, and table II herein, as specified (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 3439

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TABLE I. Type designation and chemical composition (weight percent). [1/]

Type	Common name	Copper including silver	Zinc	Tin	Manganese	Iron	Silicon	Nickel including cobalt	Phosphorus	Aluminum	Lead	Total other elements
MIL-CuSi	Copper-silicon alloy (silicon bronze)	Remainder	* [2/]	1.50 [3/]	1.50 [3/]	0.50	2.80 to 4.00	* [2/]	* [2/]	0.010* [2/]	0.020* [2/]	0.50
MIL-CuSn-C	Copper-tin alloy (phosphor bronze)	Remainder	* [2/]	7.00 to 9.00	* [2/]	* [2/]	* [2/]	* [2/]	0.10 to 0.35	0.010* [2/]	0.020* [2/]	0.50
MIL-CuAl-A2	Copper-aluminum alloy (aluminum bronze)	Remainder	0.020	--	--	0.75 to 1.50	0.10	--	--	9.0 to 11.0	0.020	0.50
MIL-CuMnNiAl	Copper-manganese-nickel-aluminum alloy (manganese, nickel, aluminum bronze)	Remainder	* [2/]	* [2/]	11.0 14.0	2.00 to 4.00	* [2/]	1.50 to 3.00	* [2/]	7.00 to 8.50	0.020	0.50
MIL-CuNiAl	Copper-nickel-aluminum alloy (nickel, aluminum bronze)	Remainder	* [2/]	* [2/]	0.60 3.50	3.00 to 5.00	* [2/]	4.00 to 5.50	* [2/]	8.50 to 9.50	0.020	0.50

NOTES:

[1/] Unless otherwise specified, single values are maximum.

[2/] See 3.3.

[3/] One or both of these elements may be present within the limits specified.

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TABLE II. Form, size, and weight.

Form	Electrode diameter (inch)	Weight (pounds) 1/
3a	All	1½ and 2½ 2/
3b	All	10 or 15 2/
3b	All	25 or 30 2/
3c	All	25 or 30 2/
3d	All	25 or 30 2/
4	All	As specified (see 6.2)
6	All	10 or 50 2/

NOTES:

1/ Tolerance on net weight is ±10 percent.

2/ Weights may be as otherwise specified (see 6.2).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-C-450 - Copper-Aluminum Alloy (Aluminum Bronze) Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 606, 610, 613, 614, and 630)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

MIL-DTL-23765 - Electrodes and Rods - Welding, Bare, Solid and Alloy-Cored, General Specification for

MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service

(Copies of these publications are available online at https://quicksearch.dla.mil.)

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2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A5.32M/A5.32 - Welding Consumables – Gases and Gas Mixtures for Fusion Welding and Allied Processes

(Copies of this publication are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM B96/B96M - Standard Specification for Copper-Silicon Alloy Plate, Sheet, Strip, and Rolled Bar for General Purposes and Pressure Vessels

ASTM B103/B103M - Standard Specification for Phosphor Bronze Plate, Sheet, Strip, and Rolled Bar

ASTM B139/B139M - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B169/B169M - Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar

ASTM E10 - Standard Test Method for Brinell Hardness of Metallic Materials

(Copies of these publications are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Electrodes and rods furnished under this specification shall be in accordance with MIL-DTL-23765 and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Chemical composition. The chemical composition of the bare electrode or rod shall be as specified in table I. Analysis shall be made for the elements indicated by an asterisk (*) in table I and those for which specific values are shown in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine their content. The total of these other elements and the elements indicated by an asterisk (*) shall not be present in excess of the limits specified for the “total other elements” in the last column of table I.

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3.4 Mechanical properties. The mechanical properties of deposited weld metal shall be as specified in table III.

TABLE III. Mechanical properties of deposited weld metal.

Type	Ultimate tensile strength (minimum, pounds per square inch [psi])	Brinell hardness number (standard ball, 3000 kg load)	Transverse side bend
MIL-CuSi	50,000	--	1/
MIL-CuSn-C	40,000	--	1/
MIL-CuAl-A2	75,000	130 to 170	--
MIL-CuMnNiAl	90,000	--	1/
MIL-CuNiAl	85,000	--	1/

NOTE:

1/ The convex surface of the specimen after bending shall have no visual cracks exceeding ⅛ inch in length. Cracks occurring on corners of a specimen during testing shall not be considered.

3.5 Cast and helix. The cast and helix of spooled and coiled electrodes shall be as specified in tables IV and V.

TABLE IV. Cast requirements.

Form (all alloy types)	Wire diameter (inch)	Cast	
		Minimum (inches)	Maximum (inches)
3a	All	6	15
3b	0.030	7	17
	0.035	10	20
	0.045	15	25
	1/16 and over	20	40
3c and 3d	All	1/	2/
4	All	3/	

NOTES:

1/ Not less than the original inside coil diameter.

2/ Not greater than 2.5 times original inside coil diameter.

3/ The cast shall be such that the filler metal will feed in an uninterrupted manner in automatic and semi-automatic equipment.

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TABLE V. Helix requirements.

Alloy type (all diameters)	Form	Helix (inches, maximum) 1/
MIL-CuSi	All	10
MIL-CuSn-C	All	
MIL-CuMnNiAl	All	6
MIL-CuNiAl	All	
MIL-CuAl-A2	All	4
All	4	2/

NOTES:

1/ Vertical separation between overlapped ends.

2/ The helix shall be such that the filler metal will feed in an uninterrupted manner in automatic and semi-automatic equipment.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on electrodes when a first article sample is required (see 3.2). This inspection shall include the examinations and the tests of table VI. Electrodes selected in accordance with MIL-DTL-23765 shall be used for the tests specified in table VI. As an alternative to using a 1/16-inch diameter electrode for first article inspection, the largest size to be produced may be used, as specified (see 6.2). When this alternative is selected, first article inspection requirements apply to at least one of the forms to be produced, as specified (see 6.2).

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TABLE VI. Summary of examinations and tests required for first article inspection.

MIL-types	Form	Examinations	Tests							Test or examination procedure	Requirements
		Visual and dimensional	Cast	Helix	Chemical	RT 1/ (soundness)	Tensile	Bend	Overlay		
All	3b, 3d, or other when specified (see 4.2)	X	--	--	--	--	--	--	--	MIL-DTL-23765	MIL-DTL-23765
		--	X	X	--	--	--	--	--	MIL-DTL-23765	3.5 (herein)
		--	--	--	X	--	--	--	--	MIL-DTL-23765 2/	3.3 (herein)
		--	--	--	--	X	--	--	--	MIL-DTL-23765 3/	MIL-DTL-23765
		--	--	--	--	--	X	X	--	MIL-DTL-23765 3/	3.4 (herein)
		--	--	--	--	--	--	--	X 4/	4.2.1 (herein)	3.4 (herein) (hardness) MIL-DTL-23765 (soundness/RT)

NOTES:

1/ "RT" is radiographic testing.

2/ In case of chemical composition dispute, the analytical procedure(s) specified in AWS A5.7/A5.7M shall be the referee method(s).

3/ Welds shall be prepared in accordance with figure 1 herein and the notes thereto.

4/ The overlay weld test applies only to type MIL-CuAl-A2.

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4.2.1 Overlay weld test. In addition to the tests specified in MIL-DTL-23765 and table VI herein, an overlay weld test shall be conducted in accordance with the following:

a. Base plate material shall be as specified in 4.4.

b. Base plate size shall be approximately ¾ by 3 by 6 inches.

c. Scale and other contaminants shall be removed from the test surface prior to welding.

d. Preheat and interpass temperature shall be as recommended by the manufacturer.

e. The overlay area approximately 1½ by 4½ inches shall be built up with layers of overlapping stringer beads to a height of ½±1/16 inch.

f. The weld surface shall be ground smooth, and the test specimen RT inspected as specified in MIL-DTL-23765.

g. Three Brinell hardness determinations shall be made on the weld metal surface in accordance with ASTM E10 for compliance with table III.

4.3 Conformance inspection. Conformance inspection shall include the examinations of MIL-DTL-23765 and the tests of table VII.

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TABLE VII. Summary of tests required for conformance inspection.

MIL-types	Form	Sizes	Cast	Helix	Chemical	RT	Tensile	Test procedure	Requirements
All	All	All	--	--	--	--	--	MIL-DTL-23765	MIL-DTL-23765
			X 1/	X 1/	--	--	--	MIL-DTL-23765	3.5 (herein)
			--	--	X	--	--	MIL-DTL-23765 2/	3.3 (herein)
			--	--	--	X	--	MIL-DTL-23765 3/	MIL-DTL-23765
			--	--	--	--	X	MIL-DTL-23765 3/	3.4 (herein)

NOTES:

1/ This test is not applicable to form 6.

2/ In case of chemical composition dispute, the analytical procedure(s) specified in AWS A5.7/A5.7M shall be the referee method(s).

3/ Welds shall be prepared in accordance with figure 1 herein and the notes thereto.

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4.3.1 Lot. For the purpose of sampling and inspection, a lot of electrodes or rods shall be the quantity of one type of alloy produced as specified in MIL-DTL-23765 or in 4.3.1.1 or 4.3.1.2 herein for bare solid electrodes.

4.3.1.1 Lot identification by heats of metal. When manufacturing processes are such that the rod is identified by a code traceable to specific heats of metal, the lot of electrodes or rods being tested shall consist of 2,000 pounds or less of one alloy, size, and form produced from a single extrusion lot of chemically conforming heats of mill coils.

4.3.1.2 Lot identification by controlled wire chemical composition. For mills using a continuous melt process, the lot of electrodes or rods being tested shall consist of 2,000 pounds or less of one alloy, size, and form produced from a single casting production schedule of which multiple lots of electrodes or rods may be produced and for which the following conditions apply:

a. For each production schedule, chemical analysis samples shall be taken at least every 2,000 pounds of material cast, either from the melt or from the initial cast material. If any sample in the production schedule run is found to not conform to the chemical control limits, then all the material produced from that production schedule run shall be rejected. Records of the chemical sampling by the mill shall be made available to the wire manufacturer, if different, and to the Government representative.

b. Intermediate product forms for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mix-ups.

c. Each mill-coiled rod produced using this lot identification method shall be chemically analyzed by approved methods.

d. Chemical analyses of the mill-coiled rod shall be certified by the laboratory and made available to the Government representative.

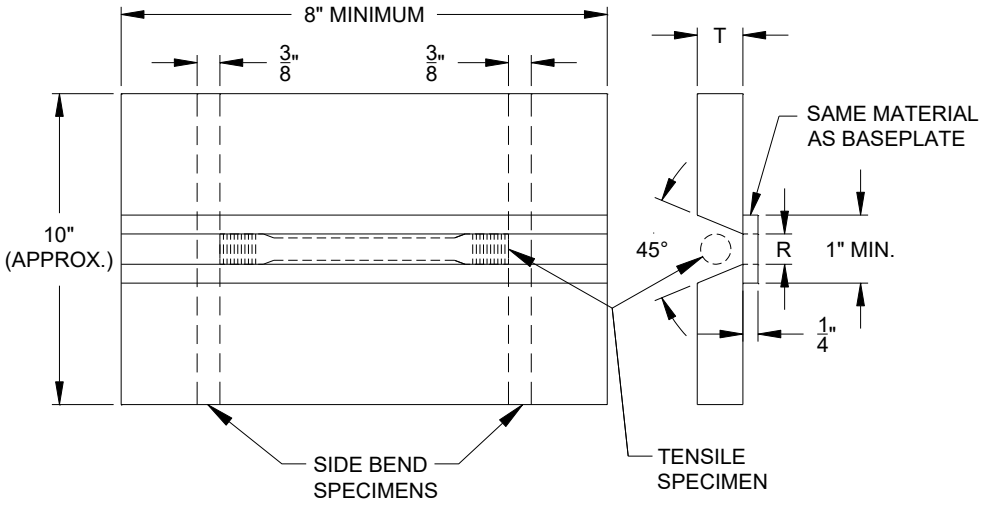
e. Mill-coiled rod conforming to established wire chemistry control for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mix-ups.

4.3.2 Sampling. Electrodes or rods, as applicable, selected for conformance inspection in accordance with MIL-DTL-23765 shall be used for the tests specified in table VII.

4.3.3 Certification of conformance. The certification of conformance shall be in accordance with MIL-DTL-23765.

4.3.3.1 Lot identification method. The lot identification method used in production of each lot shall be provided (i.e., heat of metal or controlled wire chemical composition in accordance with MIL-DTL-23765 [including revision] or MIL-DTL-23765/3 [including revision]).

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NOTES:

1. Dimensions shall be as specified in table VIII.
2. Base plate shall be as specified in table IX.
3. Welding shall be performed in the flat position.
4. The preheat and interpass temperature shall be 450 °F maximum. Water quenching between passes shall not be used.
5. For the GMAW process, direct current (DC) reverse polarity shall be used. For GTAW, DC straight polarity shall be used. Weld layers shall be made of stringer beads having a maximum thickness of 3/16 inch.
6. Arc shielding shall be helium or argon in accordance with AWS A5.32M/A5.32 or mixtures of the two gases.
7. After completion of the weld, it shall be allowed to cool, the weld reinforcement and backing strip shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the required nondestructive inspection. The weld test assembly shall be cut as shown resulting in a tensile coupon (i.e., blanks) suitable for 0.500-inch diameter specimens (or 0.250-inch diameter specimens when ½-inch base plate thickness is used) and two side bend coupons. The side bend coupons shall be used only for the first article inspection. No base metal shall be removed within ½ inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used.
8. The tensile coupon shall be machined into a 0.500-inch tensile specimen (0.250-inch diameter specimen when ½-inch plate thickness is used). Two-side bend specimens shall be prepared and tested as specified in MIL-DTL-23765.
9. Grinding (or burring) during welding of a test plate shall be limited to grinding of weld starts and grinding to correct operator error. Grinding of weld starts shall be limited to the immediate start area (first ½-inch length of weld bead deposit maximum) only and shall be done only when considered necessary by the welder. Grinding to correct operator error shall be limited to a maximum of 1 inch of weld bead length for a test plate. The amount of grinding employed (weld start and operator error) for each test plate shall be recorded by the welder on the test plate work sheet record. The record shall indicate the total number of weld starts that were ground and the length and location by weld layer of any grinding employed to correct operator error conditions.

FIGURE 1. Weld joint for soundness and for mechanical tests.

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TABLE VIII. Weld test details.

Test classification	Form	Type	Wire, rod diameter (inch)	T (min) (inch)	R [1/] (inch)	Tensile specimen (inch, diameter)	Welding process
First article inspection	3b or 3d, or other when specified (see 4.2)	All	1/16	3/4	1/2	0.500	GMAW
Conformance inspection	3a, 3b, 3c, 3d, and 4	All	Less than 1/16	1/2	1/4	0.250	
			1/16 and larger	3/4	1/2	0.500	
	6	All	All	3/4	1/2	0.500	GTAW

NOTE:

[1/] Root opening "R" tolerance shall be ±1/16 inch.

4.4 Base metals for groove weld test plates. The base metals for the groove weld test plates (including the backing straps) shall be as specified in table IX. MIL-S-22698 grade B shall be used for overlay welds.

TABLE IX. Base metals for groove weld test plates.

Electrode type	Plate metal composition	Document [1/]
MIL-CuSi	Copper alloy number C65500	ASTM B96
MIL-CuSn-C	Copper alloy number C51000	ASTM B139 or ASTM B103
MIL-CuAl-A2	Copper alloy number C61400	QQ-C-450 or ASTM B169
MIL-CuMnNiAl	Copper alloy number C95700	MIL-B-24480/ASTM B148
MIL-CuNiAl	Copper alloy numbers C95700 or C63000	MIL-B-24480/ASTM B148 or ASTM B150

NOTE:

[1/] Any applicable procurement specification meeting the designated UNS chemistry composition may be used.

4.5 Inspection of packaging and marking. Inspection of packaging and marking shall be in accordance with MIL-DTL-23765.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by Department of Defense (DoD) or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The copper and copper alloy solid bare welding electrodes and rods covered by this specification are capable of depositing sound weld metal that meets the mechanical properties specified herein when welding on similar composition base metal using the GMAW process with DC, reversed polarity, or the GTAW process with DC, straight polarity, as applicable. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.1.1 Type MIL-CuSi. Type MIL-CuSi is intended for use in welding copper-silicon alloys (silicon bronze).

6.1.2 Type MIL-CuSn-C. Type MIL-CuSn-C is intended for use in welding copper-tin alloys (phosphor bronze).

6.1.3 Type MIL-CuAl-A2. Type Mil-CuAl-A2, an iron bearing alloy, is intended for use in joining copper-aluminum alloys (aluminum bronze), copper-silicon alloys (silicon bronze), many ferrous metals and alloys, and combinations of dissimilar metals. This type is also suitable for surfacing bearing, wear-resistant and corrosion-resistant surfaces.

6.1.4 Type MIL-CuMnNiAl. Type MIL-CuMnNiAl is intended for use in welding copper-manganese-nickel-aluminum alloy (manganese, nickel, aluminum bronze) castings.

6.1.5 Type MIL-CuNiAl. Type MIL-CuNiAl is intended for use in welding copper-nickel-aluminum (nickel, aluminum bronze) castings in accordance with MIL-B-24480.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Type, form, and size required (see 1.2).
c. Weight required (see table II).
d. When first article is required (see 3.2).
e. For first article testing, the largest electrode diameter to be tested as an alternate to 1/16-in diameter (see 4.2).
f. For first article testing, the form(s) to be tested (see 4.2).
g. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Aluminum bronze
First article inspection
Gas metal-arc welding (GMAW)
Gas tungsten-arc welding (GTAW)
Manganese nickel aluminum bronze
Nickel aluminum bronze
Phosphor bronze
Silicon bronze

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
(Project 3439-2024-061)

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